What is Container Runtime?

Professional Answer

The container runtime is responsible for pulling images, creating containers, starting, and stopping them according to Pod specifications.

Unprofessional Answer

Docker runs containers.

What is a container runtime in Kubernetes?

The container runtime is responsible for running containers on worker nodes.
It pulls container images and starts, stops, and manages container processes.
Kubernetes communicates with the runtime using the Container Runtime Interface (CRI).
Common runtimes include containerd and CRI-O.
This abstraction allows Kubernetes to be independent of any specific runtime.

Container Runtime (CRI Internals)

What it is

- Implements **Container Runtime Interface**
- Examples:
    - containerd
    - CRI-O

Internal Steps

1. Pull image
2. Create container
3. Create namespaces & cgroups
4. Start container process
5. Monitor container lifecycle

What is a container runtime?

A container runtime is the **software responsible for running containers on a node**.
It pulls images, creates containers, starts and stops them.
It manages namespaces, cgroups, and filesystem isolation.
Kubernetes does **not** talk to the runtime directly.
It communicates through **CRI (Container Runtime Interface)**.

Why container runtime is important?

- Executes Pods on worker nodes
- Enforces resource limits
- Handles container lifecycle
- Interfaces with OS kernel
- Without it, Pods cannot run

Runtime = execution engine.

CRI (Container Runtime Interface)

CRI is a **standard API** defined by Kubernetes.
It allows Kubernetes to support **multiple runtimes**.
kubelet uses CRI to talk to the runtime.
This removed Kubernetes' dependency on Docker.
CRI enables pluggable runtimes.

kubelet ↔ container runtime flow

1. kubelet receives Pod spec
2. kubelet calls CRI
3. Runtime pulls image
4. Runtime creates container
5. Container starts
6. Status returned to kubelet

kubelet never executes containers itself.

What are the popular container runtimes?

1. containerd (Most common)

- CNCF project
- Used by Docker internally
- Lightweight
- Production standard

2. CRI-O

- Kubernetes-native runtime
- Used by OpenShift

- Minimal and secure

3. Docker (Deprecated)

- Removed in Kubernetes 1.24+
- Docker Shim removed
- Still used as image builder

What happens if container runtime goes down?

- Running containers keep running
- New Pods fail to start
- kubelet reports errors
- Node becomes NotReady
- Requires runtime restart